AI301
Computer System Architecture
Lab1

Contents

1. Concepts about types of computers
2. Program running flow
3. Simple calculation about storage
4. CPI
5. Integrated Circuit cost
6. Simple calculation about parallel computation

Question 1

Q1. Aside from the smart cell phones used by a billion people, list and describe four other types of computers.

Personal computer (includes workstation and laptop): Personal computers emphasize delivery of good performance to single users at low cost and usually execute third-party software.

Personal mobile device (PMD, includes tablets): PMDs are battery operated with wireless connectivity to the Internet and typically cost hundreds of dollars, and, like PCs, users can download software ("apps") to run on them. Unlike PCs, they no longer have a keyboard and mouse, and are more likely to rely on a touch-sensitive screen or even speech input.

Server: Computer used to run large problems and usually accessed via a network.

Question 1

Warehouse scale computer: Thousands of processors forming a large cluster.

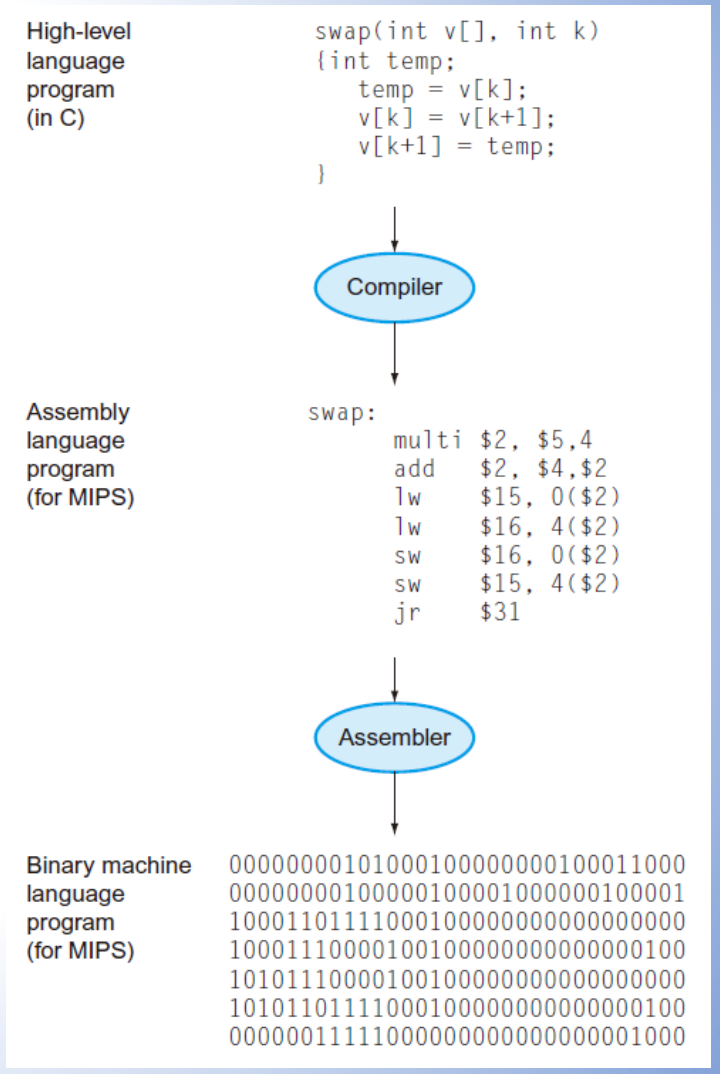
Supercomputer: Computer composed of hundreds to thousands of processors and terabytes of memory.

Embedded computer: Computer designed to run one application or one set of related applications and integrated into a single system.

Question 2

Q2. Describe the steps that transform a program written in a high-level language such as C into a representation that is directly executed by a computer processor.

The program is compiled into an assembly language program, which is then assembled into a machine language program.



Question 3

Q3. Assume a color display using 8 bits for each of the primary colors (red, green, blue) per pixel and a frame size of 1280 × 1024.

a. What is the minimum size in bytes of the frame buffer to store a frame?

b. How long would it take, at a minimum, for the frame to be sent over a 100 Mbit/s network?

Question 3

Q3. Assume a color display using 8 bits for each of the primary colors (red, green, blue) per pixel and a frame size of 1280 × 1024.

a. What is the minimum size in bytes of the frame buffer to store a frame?

b. How long would it take, at a minimum, for the frame to be sent over a 100 Mbit/s network?

a. 1280 × 1024 pixels = 1,310,720 pixels => 1,310,720 × 3 = 3,932,160 bytes/frame.

b. 3,932,160 bytes × (8 bits/byte) /100E6 bits/second = 0.31 seconds

Metric prefix

List of SI prefixes [edit]

The BIPM specifies twenty prefixes for the International System of Units (SI).

SI prefixes V • T • E

Prefix		Base 1000	Base 10	Decimal	English word		Adoption[nb 1]
Name	Symbol				Short scale	Long scale	
yotta	Y	1000^8	10^{24}	1 000 000 000 000 000 000 000 000	septillion	quadrillion	1991
zetta	Z	1000^7	10^{21}	1 000 000 000 000 000 000 000	sextillion	trilliard	1991
exa	E	1000^6	10^{18}	1 000 000 000 000 000 000	quintillion	trillion	1975
peta	P	1000^5	10^{15}	1 000 000 000 000 000	quadrillion	billiard	1975
tera	T	1000^4	10^{12}	1 000 000 000 000	trillion	billion	1960
giga	G	1000^3	10^9	1 000 000 000	billion	milliard	1960
mega	M	1000^2	10^6	1 000 000	million		1873
kilo	k	1000^1	10^3	1 000	thousand		1795

CPI

CPI: Cycles per instruction, is one aspect of a processor's performance: the average number of clock cycles per instruction for a program or program fragment

Example 1 [edit]

For the multi-cycle MIPS, there are five types of instructions:

- Load (5 cycles)
- Store (4 cycles)
- R-type (4 cycles)
- Branch (3 cycles)
- Jump (3 cycles)

If a program has:

- 50% load instructions
- 25% store instructions
- 15% R-type instructions
- 8% branch instructions
- 2% jump instructions

then, the CPI is:

$$\mathrm{CPI} = \frac{5 \times 50 + 4 \times 25 + 4 \times 15 + 3 \times 8 + 3 \times 2}{100} = 4.4$$

Question 4

Comparing Code Segments

A compiler designer is trying to decide between two code sequences for a particular computer. The hardware designers have supplied the following facts:

	CPI for each instruction class		
	A	B	C
CPI	1	2	3

For a particular high-level language statement, the compiler writer is considering two code sequences that require the following instruction counts:

Code sequence	Instruction counts for each instruction class		
	A	B	C
1	2	1	2
2	4	1	1

Which code sequence executes the most instructions? Which will be faster? What is the CPI for each sequence?

Question 4

Sequence 1 executes 2 + 1 + 2 = 5 instructions. Sequence 2 executes 4 + 1 + 1 = 6 instructions. Therefore, sequence 1 executes fewer instructions.

We can use the equation for CPU clock cycles based on instruction count and CPI to find the total number of clock cycles for each sequence:

$$\text{CPU clock cycles} = \sum_{i=1}^{n} (\text{CPI}_i \times \text{C}_i)$$

This yields

$$\text{CPU clock cycles}_1 = (2 \times 1) + (1 \times 2) + (2 \times 3) = 2 + 2 + 6 = 10 \text{ cycles}$$

$$\text{CPU clock cycles}_2 = (4 \times 1) + (1 \times 2) + (1 \times 3) = 4 + 2 + 3 = 9 \text{ cycles}$$

So code sequence 2 is faster, even though it executes one extra instruction. Since code sequence 2 takes fewer overall clock cycles but has more instructions, it must have a lower CPI. The CPI values can be computed by

$$\text{CPI} = \frac{\text{CPU clock cycles}}{\text{Instruction count}}$$

$$\text{CPI}_1 = \frac{\text{CPU clock cycles}_1}{\text{Instruction count}_1} = \frac{10}{5} = 2.0$$

$$\text{CPI}_2 = \frac{\text{CPU clock cycles}_2}{\text{Instruction count}_2} = \frac{9}{6} = 1.5$$

Question 5

Consider three different processors P1, P2, and P3 executing the same instruction set. P1 has a 3 GHz clock rate and a CPI of 1.5. P2 has a 2.5 GHz clock rate and a CPI of 1.0. P3 has a 4.0 GHz clock rate and has a CPI of 2.2.

a Which processor has the highest performance expressed in instructions per second?

b If the processors each execute a program in 10 seconds, find the number of cycles and the number of instructions.

Question 5

Consider three different processors P1, P2, and P3 executing the same instruction set. P1 has a 3 GHz clock rate and a CPI of 1.5. P2 has a 2.5 GHz clock rate and a CPI of 1.0. P3 has a 4.0 GHz clock rate and has a CPI of 2.2.

a. Which processor has the highest performance expressed in instructions per second?

a. performance of P1 (instructions/sec) $= 3 \times 10^9/1.5 = 2 \times 10^9$

performance of P2 (instructions/sec) $= 2.5 \times 10^9/1.0 = 2.5 \times 10^9$

performance of P3 (instructions/sec) $= 4 \times 10^9/2.2 = 1.8 \times 10^9$

b. If the processors each execute a program in 10 seconds, find the number of cycles and the number of instructions.

b. cycles(P1) $= 10 \times 3 \times 10^9 = 30 \times 10^9$ s

cycles(P2) $= 10 \times 2.5 \times 10^9 = 25 \times 10^9$ s

cycles(P3) $= 10 \times 4 \times 10^9 = 40 \times 10^9$ s

No. instructions(P1) $= 30 \times 10^9/1.5 = 20 \times 10^9$

No. instructions(P2) $= 25 \times 10^9/1 = 25 \times 10^9$

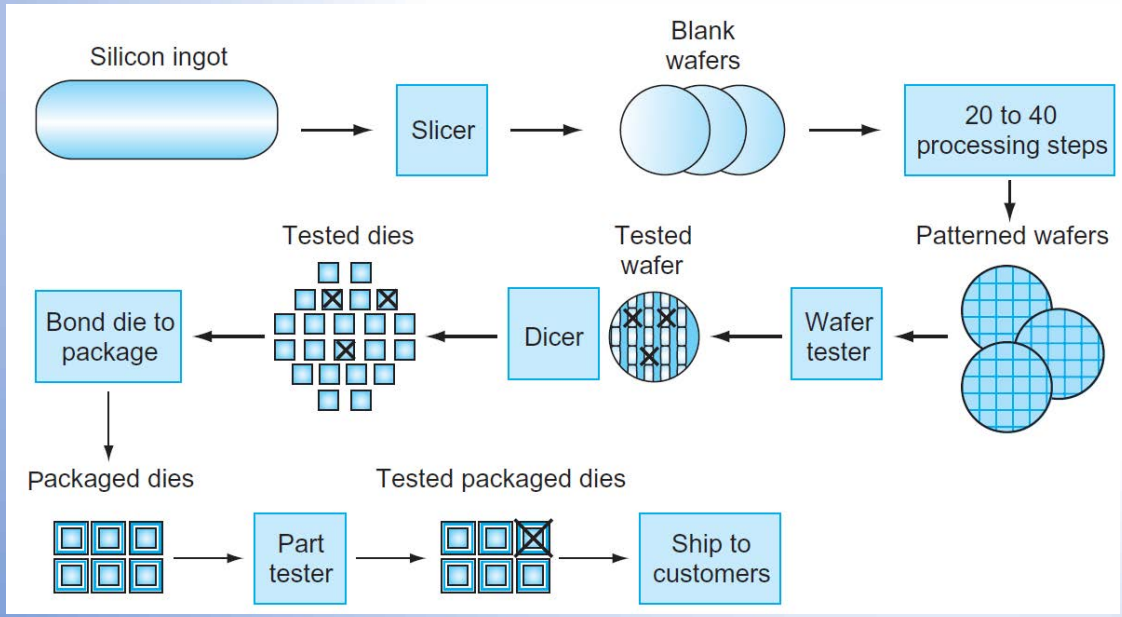
No. instructions(P3) $= 40 \times 10^9/2.2 = 18.18 \times 10^9$

Integrated Circuit cost

Defect: A microscopic flaw in a wafer or in patterning steps that can result in the failure of the die containing that defect.

Die: The individual rectangular sections that are cut from a wafer, more informally known as chips.

Yield: The percentage of good dies from the total number of dies on the wafer.



$$\text{Cost per die} = \frac{\text{Cost per wafer}}{\text{Die per wafer} \times \text{yield}}$$

$$\text{Dies per wafer} \approx \frac{\text{Wafer area}}{\text{Die area}}$$

$$\text{Yield} = \frac{1}{[1+(\text{Defects per area} \times \text{Die area}/2)]^2}$$

Question 6

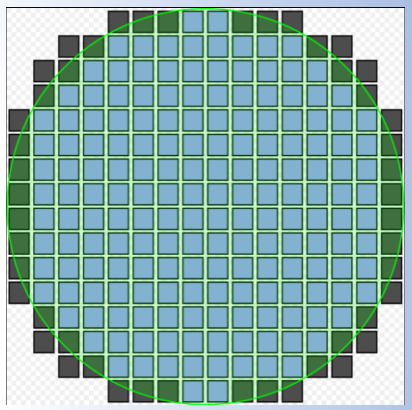
Q6. Assume a 15 cm diameter wafer has a cost of 12, contains 84 dies, and has 0.020 defects/cm 2 . Assume a 20 cm diameter wafer has a cost of 15, contains 100 dies, and has 0.031 defects/cm 2 .

a. Find the yield for both wafers.
b. Find the cost per die for both wafers.
c. If the number of dies per wafer is increased by 10% and the defects per area unit increases by 15%, find the die area and yield.
d. Assume a fabrication process improves the yield from 0.92 to 0.95. Find the defects per area unit for each version of the technology given a die area of 200 mm 2 .

Question 6

Q6. Assume a 15 cm diameter wafer has a cost of 12, contains 84 dies, and has 0.020 defects/cm 2 . Assume a 20 cm diameter wafer has a cost of 15, contains 100 dies, and has 0.031 defects/cm 2 .

a. Find the yield for both wafers.



$$\text{die area}_{15\text{cm}} = \text{wafer area/dies per wafer} = \text{pi}*7.5^2 / 84 = 2.10\ \text{cm}^2$$

$$\text{yield}_{15\text{cm}} = 1/(1+(0.020*2.10/2))^2 = 0.9593$$

$$\text{die area}_{20\text{cm}} = \text{wafer area/dies per wafer} = \text{pi}*10^2/100 = 3.14\ \text{cm}^2$$

$$\text{yield}_{20\text{cm}} = 1/(1+(0.031*3.14/2))^2 = 0.9093$$

b. Find the cost per die for both wafers.

$$\text{cost/die}_{15\text{cm}} = 12/(84*0.9593) = 0.1489$$

$$\text{cost/die}_{20\text{cm}} = 15/(100*0.9093) = 0.1650$$

$$\text{Cost per die} = \frac{\text{Cost per wafer}}{\text{Die per wafer} \times \text{yield}}$$

$$\text{Dies per wafer} \approx \frac{\text{Wafer area}}{\text{Die area}}$$

$$\text{Yield} = \frac{1}{[1+(\text{Defects per area} \times \text{Die area}/2)]^2}$$

Question 6

Q6. Assume a 15 cm diameter wafer has a cost of 12, contains 84 dies, and has 0.020 defects/cm 2 . Assume a 20 cm diameter wafer has a cost of 15, contains 100 dies, and has 0.031 defects/cm 2 .

c. If the number of dies per wafer is increased by 10% and the defects per area unit increases by 15%, find the die area and yield.

$$\text{die area}_{15\text{cm}} = \text{wafer area/dies per wafer} = \text{pi}*7.5^2/(84*1.1) = 1.91\ \text{cm}^2$$

$$\text{yield}_{15\text{cm}} = 1/(1 + (0.020*1.15*1.91/2))^2 = 0.9575$$

$$\text{die area}_{20\text{cm}} = \text{wafer area/dies per wafer} = \text{pi}*10^2/(100*1.1) = 2.86\ \text{cm}^2$$

$$\text{yield}_{20\text{cm}} = 1/(1 + (0.03*1.15*2.86/2))^2 = 0.9082$$

d. Assume a fabrication process improves the yield from 0.92 to 0.95. Find the defects per area unit for each version of the technology given a die area of 200 mm 2 .

$$\text{defects per area}_{0.92} = (1-\text{y}\text{\^{}}.5)/(\text{y}\text{\^{}}.5*\text{die_area}/2) = (1-0.92\text{\^{}}.5)/(0.92\text{\^{}}.5*2/2) = 0.043\ \text{defects/cm}^2$$

$$\text{defects per area}_{0.95} = (1-\text{y}\text{\^{}}.5)/(\text{y}\text{\^{}}.5*\text{die_area}/2) = (1-0.95\text{\^{}}.5)/(0.95\text{\^{}}.5*2/2) = 0.026\ \text{defects/cm}^2$$

Question 7

Q7. When a program is adapted to run on multiple processors in a multiprocessor system, the execution time on each processor is comprised of computing time and the overhead time required for locked critical sections and/or to send data from one processor to another.
Assume a program requires t = 100 s of execution time on one processor. When run p processors, each processor requires t/p s, as well as an additional 4 s of overhead, irrespective of the number of processors. Compute the per-processor execution time for 2, 4, 8, 16, 32, 64, and 128 processors. For each case, list the corresponding speedup relative to a single processor and the ratio between actual speedup versus ideal speedup (speedup if there was no overhead).

Question 7

Q7. When a program is adapted to run on multiple processors in a multiprocessor system, the execution time on each processor is comprised of computing time and the overhead time required for locked critical sections and/or to send data from one processor to another.

Assume a program requires t = 100 s of execution time on one processor. When run p processors, each processor requires t/p s, as well as an additional 4 s of overhead, irrespective of the number of processors. Compute the per-processor execution time for 2, 4, 8, 16, 32, 64, and 128 processors. For each case, list the corresponding speedup relative to a single processor and the ratio between actual speedup versus ideal speedup (speedup if there was no overhead).

processors	exec. time/ processor	time w/overhead	speedup	actual speedup/ideal speedup
1	100			
2	50	54	100/54 = 1.85	1.85/2 = .93
4	25	29	100/29 = 3.44	3.44/4 = 0.86
8	12.5	16.5	100/16.5 = 6.06	6.06/8 = 0.75
16	6.25	10.25	100/10.25 = 9.76	9.76/16 = 0.61